



HDI / Microvia PCBs

High-Density Interconnect — building modern mobile + dense BGAs

When standard vias and 6-mil traces aren't dense enough

Audience: Engineers working with fine-pitch BGAs

Build-up • Microvias • Stacked vs staggered • Cost • When to use



What is HDI

- ▶ HDI = High-Density Interconnect — fine-feature PCB technology
- ▶ Defined by IPC: features finer than standard PCB capability
- ▶ Trace/space: 50 μm / 50 μm (2 mil / 2 mil) — vs standard 100 μm / 100 μm
- ▶ Microvia: 50-150 μm diameter — vs standard 0.3 mm
- ▶ Build-up construction: multiple sequential laminations
- ▶ Used in: smartphones, wearables, dense BGAs (200+ pins on small board)
- ▶ Cost: 2-5 $\times$ standard PCB
- ▶ Lead time: 2-4 $\times$ standard

Microvia Basics

- ▶ Microvia: small via (50-150 μm diameter) drilled by LASER, not mechanical
- ▶ Blind microvia: drilled from outer layer down to first inner layer only
- ▶ Buried microvia: between two inner layers (formed before outer-layer lamination)
- ▶ Filled microvia: plated solid with copper (improves SI, allows via-in-pad)
- ▶ Stacked microvia: vertical stack of microvias for direct layer-to-layer connection
- ▶ Staggered microvia: offset adjacent microvias (mechanically stronger than stacked)
- ▶ Aspect ratio (depth / diameter): laser microvias usually $< 1:1$ (shallow)

HDI Build-up Construction

- ▶ Build-up = sequential lamination — each layer added one at a time
- ▶ Type 1 HDI: $1+N+1$ (one build-up layer on top of N-layer core)
- ▶ Type 2 HDI: $2+N+2$ (two build-up layers)
- ▶ Type 3 HDI: $3+N+3$ (three build-up layers)
- ▶ Any-layer HDI: every layer can host microvias
- ▶ Each build-up step adds: image, drill, plate, etch, laminate
- ▶ Cost scales linearly with build-up count
- ▶ Most consumer products use Type 1 or 2 HDI

Stacked vs Staggered Microvias

Stacked microvias

- ▶ Vertical stack of microvias
- ▶ Direct path through multiple layers
- ▶ Shortest possible interconnect
- ▶ Requires fill (cap-via)
- ▶ Higher cost (extra plating + fill)
- ▶ Some reliability concerns (thermal stress)

Staggered microvias

- ▶ Offset adjacent microvias
- ▶ Slightly longer signal path
- ▶ Mechanically stronger
- ▶ Preferred by some fabs
- ▶ Less expensive than stacked
- ▶ Slightly more board area required

When to Use HDI

- ▶ BGA with > 200 balls AND pitch < 0.65 mm — standard escape impossible
- ▶ Board area constrained (mobile, wearable, dense IoT)
- ▶ Mixed-signal product with strict ground separation
- ▶ When standard PCB requires too many layers (HDI replaces 4-6 layers with 2 microvia builds)
- ▶ Multi-die packaging (system-in-package, chip stacks)
- ▶ Smartphone-class designs (Apple A-series, Snapdragon)
- ▶ When cost is NOT a primary driver

Standard vs HDI Capability Comparison



Spec	Standard PCB	HDI PCB
Min trace width	100 µm (4 mil)	50 µm (2 mil)
Min trace spacing	100 µm (4 mil)	50 µm (2 mil)
Min via diameter	300 µm (12 mil)	75-150 µm laser
Min annular ring	100 µm (4 mil)	25 µm (1 mil)
Aspect ratio	10:1	1:1 microvia
Layer count	1-32	Typically 6-24 + microvia layers
Cost vs standard	1.0×	2-5×

BGA Escape with HDI

- ▶ Standard escape: dog-bone (pad on top + via beside it) — needs 0.5 mm pitch min
- ▶ HDI escape: via-in-pad (microvia directly under each ball)
- ▶ Via-in-pad requires FILLED + PLATED microvias (or solder wicks into the via)
- ▶ Microvia diameter 100 μm fits inside 0.4 mm pitch BGA pad
- ▶ Inner-layer routing freed up — single layer can break out 200+ ball BGA
- ▶ Combined with stacked microvias: any-layer access
- ▶ Smartphone SoCs (Apple, Qualcomm) all use HDI for BGA escape

HDI Materials

- ▶ Standard FR-4 OK for non-RF HDI ($T_g \geq 170$)
- ▶ Megtron 6 / 7 for high-speed HDI (PCIe Gen 4+, DDR4/5)
- ▶ Specialty: BT resin (for IC substrates)
- ▶ Thinner cores (50-100 μm) than standard FR-4
- ▶ Build-up dielectric typically RCC (Resin-Coated Copper) or AjinomotoBF (ABF)
- ▶ Surface finish: ENIG essential (flat for fine-pitch BGA)
- ▶ All multilayer rules apply, just at smaller dimensions

HDI Design Guidelines

- ▶ Talk to fab early — HDI capability varies dramatically
- ▶ Use vendor design rules (each HDI fab has unique trace/space/via specs)
- ▶ Microvia diameter typically 100-150 μm ; aspect ratio $< 1:1$
- ▶ Microvia plating thickness 15-25 μm
- ▶ Annular ring 25 μm minimum (vs 100 μm standard)
- ▶ Via-in-pad: confirm fab does filling + plating (not all do)
- ▶ Layer-pair routing: organise stackup so signals can use microvias optimally
- ▶ Stacking: confirm vendor reliability data — some patterns fail thermal cycling

Cost Drivers

- ▶ Number of microvia layers (Type 1 cheapest, Type 3+ premium)
- ▶ Build-up count (1 build-up = 1.5×, 2 build-ups = 2-2.5×, etc.)
- ▶ Via fill (capped microvias) adds 20-30% cost
- ▶ Stacked vs staggered (stacked slightly more expensive)
- ▶ Material — Megtron 6 vs FR-4 doubles material cost
- ▶ Tighter trace/space (sub-50 μm) → premium fabs only, 3-5× cost
- ▶ Volume — HDI cost drops less than standard at volume

HDI Fab Sources

- ▶ Premium HDI: Unimicron (Taiwan), AT&S (Austria + India), Ibiden (Japan)
- ▶ Mid-tier HDI: PCB Way premium tier, JLCPCB advanced
- ▶ Mass-market HDI: not really 'cheap' anywhere — minimum bar is high
- ▶ Lead times: 15-30 days typical (vs 5-10 for standard)
- ▶ Indian capability: AT&S India + Centum offer HDI; verify capability
- ▶ For prototype quantities (5-10 boards) — many smaller HDI fabs offer expensive but fast service

Key Takeaways

Key takeaways from this lecture

HDI = density tech

Sub-100 μm features.

Microvias enable BGA escape

Via-in-pad for 0.4 mm pitch.

Build-up adds layers

Type 1 / 2 / 3 by build-up count.

Cost 2-5 $\times$ standard

Plan accordingly.

Talk to fab early

Capability varies widely.

ENIG required

For fine-pitch BGA flatness.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ IPC-2226 — Sectional Design Standard for HDI Printed Boards
- ▶ IPC-6016 — Qualification spec for HDI
- ▶ Unimicron / AT&S / Ividen — premium HDI vendors
- ▶ Phil's Lab YouTube — HDI walkthroughs
- ▶ Sierra Circuits HDI design e-book
- ▶ Cadence Allegro HDI workflow
- ▶ AT&S India — local HDI source
- ▶ DesignCon papers on HDI signal integrity

Thank you • Build something this week.